

Q6(a)

- Radiation emitted does not all enter the detector
- absorption in air before reaching detector
- detector not sensitive to all radiations
- background radiation

Any one point.

6(b)(i)

The curve is not smooth or curve fluctuates/curve is jagged .

6(b)(ii)

From graph, background count = 9 count per min.

At $t = 0$, count rate due to source alone = $170 - 9 = 161$ count per min

At half-life, count rate = $161/2 + 9 = 89.5$ count per min

From graph, half life = 1.5 hr

OR

[At $t = 1$ hr, count rate due to source alone = $116 - 9 = 107$ count per min

At half-life, count rate = $107/2 + 9 = 62.5$ count per min

From graph, half life = $2.5 - 1 = 1.5$ hr

Half-life = 1.5 hr]

6(c)(i)

There is no change in half life.

Decay is spontaneous, independent of physical factors such as temperature.

6(c)(ii)

The count rate is likely to be different.

Decay is random resulting in fluctuations in reading.

SECTION C